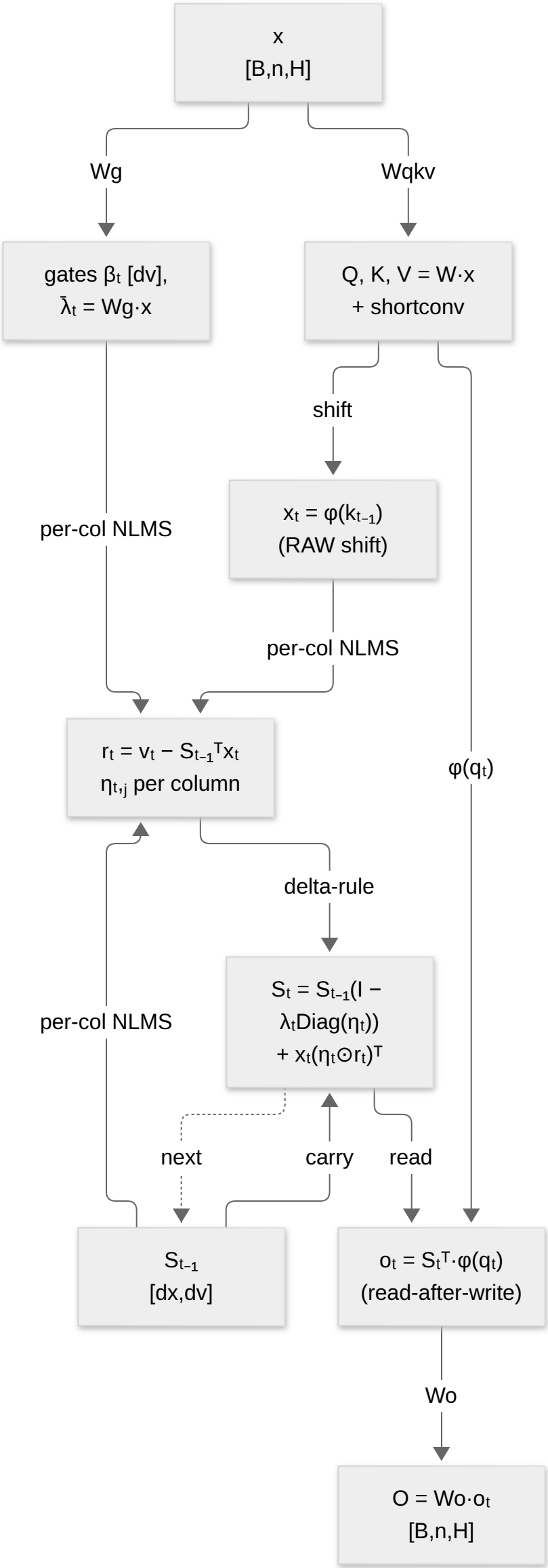

x
[B,n,H]
Wg
Wqkv
gates β_t [dv],
$\bar{\lambda}_t = Wg \cdot x$
Q, K, V = W·x
+ shortconv
shift
$x_t = \varphi(k_{t-1})$
(RAW shift)
per-col NLMS
per-col NLMS
$r_t = v_t - S_{t-1}^T x_t$
$\eta_{t,j}$ per column
$\varphi(q_t)$
delta-rule
$S_t = S_{t-1}(I - \lambda_t Diag(\eta_t)) + x_t(\eta_t \odot r_t)^T$
per-col NLMS
next
carry
read
S_{t-1}
[dx,dv]
$o_t = S_t^T \cdot \varphi(q_t)$
(read-after-write)
Wo
O = Wo·o_t
[B,n,H]